MATH 4301/5031 **Supplementary Practice Midterm 1** **Fall 2026**

Full Name: _____

Instructions

1. **Part I** (Problems 1–5) follows the format of the practice midterm, with every problem pushed further. Time yourself: 75 minutes. **Part II** (Problems 6–8 and the Bonus) goes beyond it and is untimed.
2. Answers should be completely justified unless otherwise noted. Counterexamples must be explicit, and you should check that they work.
3. Do not use determinants, eigenvalues, or inner products.
4. Notation: $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$. $\ker T$ and $\operatorname{Ran} T$ are the kernel and range of T. $\mathcal{P}_n$ is the space of real polynomials of degree at most n, and $\mathcal{P}$ is the space of all real polynomials. E_{ij} is the matrix with a 1 in entry (i, j) and 0 everywhere else.
5. Write your answers in the space provided. If you need more space, continue on the back.

Problem	1	2	3	4	5	6	7	8	Bonus	Total
Points	24	18	22	14	22	15	15	20	+10	150
Score										

Do not turn the page until you start the clock.

Part I

[24] 1. Determine whether each statement below is true or false. Briefly justify your answer with a proof or an explicit counterexample.

(a) The map $T\colon \mathbb{C} \to \mathbb{C}$, $T(z) = \overline{z}$, is a linear transformation of $\mathbb{C}$ regarded as a vector space over $\mathbb{C}$.

(b) Row operations preserve the column space of a matrix.

(c) If $v_1, v_2, v_3 \in V$ are linearly dependent, then v_3 is a linear combination of v_1 and v_2.

(d) If $A \in M_{m\times n}(\mathbb{F})$ and the equation $Ax = b$ has exactly one solution for some $b \in \mathbb{F}^m$, then $Ax = c$ has at most one solution for every $c \in \mathbb{F}^m$.

(e) If the columns of $A \in M_{m\times n}(\mathbb{F})$ are linearly independent, then A is right invertible.

(f) If X and Y are subspaces of V and $X \cup Y = V$, then $X = V$ or $Y = V$.

(g) There exists a linear map $T\colon \mathbb{R}^5 \to \mathbb{R}^5$ with $\ker T = \operatorname{Ran} T$.

(h) Every 4-dimensional subspace of $\mathbb{R}^6$ and every 3-dimensional subspace of $\mathbb{R}^6$ have a nonzero vector in common.

[18] 2. Let V be a vector space over $\mathbb{F}$.

(a) Define what it means for vectors $v_1, \ldots, v_m \in V$ to be *linearly independent*, and what it means for them to be a *basis* of V.

(b) Using only the axioms of a vector space, prove that $\alpha 0 = 0$ for every $\alpha \in \mathbb{F}$. Then prove that if $\alpha \in \mathbb{F}$ and $v \in V$ satisfy $\alpha v = 0$, then $\alpha = 0$ or $v = 0$.

(c) Let $v_1, \ldots, v_n$ be a basis of V, and for $k = 1, \ldots, n$ let $w_k = v_k + v_{k+1} + \cdots + v_n$. Prove that $w_1, \ldots, w_n$ is a basis of V.

(d) Let $n \geq 2$ and let $v_1, \ldots, v_n \in V$ be linearly independent. Determine, with proof, all n for which the vectors

$$v_1 + v_2, \quad v_2 + v_3, \quad \ldots, \quad v_{n-1} + v_n, \quad v_n + v_1$$

are linearly independent.

[22] 3.

(a) Let $U = \{x \in \mathbb{F}^6 : x_1 + x_2 + x_3 = 0 \text{ and } x_4 = x_5 = x_6\}$. Find the smallest positive integer m for which there exists a linear map $T\colon \mathbb{F}^6 \to \mathbb{F}^m$ with $\ker T = U$. Prove that no smaller m works, and write down such a T explicitly for your value of m.

(b) Let V be finite-dimensional and let $S, T\colon V \to V$ be linear. Prove that if ST is invertible, then TS is invertible.

(c) Show that the conclusion of (b) can fail when V is infinite-dimensional: find linear maps $S, T\colon \mathcal{P} \to \mathcal{P}$ such that $ST = I$ but TS is not invertible.

(d) Now let V be an arbitrary vector space, possibly infinite-dimensional, and let $S, T\colon V \to V$ be linear. Prove that if STS is invertible, then S and T are both invertible.

[14] 4. Let V be the set of all $A \in M_{3\times 3}(\mathbb{R})$ such that every row and every column of A sums to 0.

(a) Show that V is a subspace of $M_{3\times 3}(\mathbb{R})$.

(b) Find $\dim V$. Justify your answer by exhibiting a basis of V and proving that it is a basis.

(c) Let W be the set of all $A \in M_{3\times 3}(\mathbb{R})$ whose three row sums and three column sums are all equal to one another (the common value may depend on A). Prove that W is a subspace of $M_{3\times 3}(\mathbb{R})$ and that $\dim W = \dim V + 1$.

[22] 5. Let $V = \{X \in M_{2\times 2}(\mathbb{R}) : X^T = X\}$, with basis $\mathcal{B} = (E_{11},\ E_{12} + E_{21},\ E_{22})$. Let

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

and define $T\colon V \to V$ by $T(X) = MXM^T$.

(a) Show that T maps V into V and that T is linear.

(b) Find the matrix $[T]_{\mathcal{B}}$ of T with respect to $\mathcal{B}$.

(c) Show that T is invertible, and compute $[T^{-1}]_{\mathcal{B}}$.

(d) Find $X \in V$ such that $T(X) = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$.

(e) The matrices

$$C_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad C_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C_3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

form a basis $\mathcal{C}$ of V. Write down the change-of-coordinates matrix that converts $\mathcal{C}$-coordinates into $\mathcal{B}$-coordinates, and use it to compute $[T]_{\mathcal{C}}$.

Part II

[15] 6. Let $D\colon \mathcal{P}_3 \to \mathcal{P}_3$ be differentiation, $Dp = p'$, and let $T = I - D$, so that $(Tp)(t) = p(t) - p'(t)$.

(a) Write down the matrices of D and of T with respect to the basis $1, t, t^2, t^3$ of $\mathcal{P}_3$.

(b) Let $N\colon V \to V$ be a linear map on any vector space V such that $N^k = 0$ for some $k \geq 1$. Prove that $I - N$ is invertible, with

$$(I - N)^{-1} = I + N + N^2 + \cdots + N^{k-1}.$$

Deduce that T is invertible, and write down the matrix of T^{-1} with respect to $1, t, t^2, t^3$.

(c) Without row reducing anything, find the unique $p \in \mathcal{P}_3$ with $p(t) - p'(t) = t^3 - 3t^2 + 1$.

(d) Prove that $p \mapsto p - p'$ is an invertible linear map $\mathcal{P} \to \mathcal{P}$, even though differentiation $D\colon \mathcal{P} \to \mathcal{P}$ is not invertible.

[15] 7. For a square matrix $X = (x_{ij})$, let $\operatorname{trace} X = \sum_i x_{ii}$.

(a) Let $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times m}(\mathbb{F})$. Prove that $\operatorname{trace}(AB) = \operatorname{trace}(BA)$. (Note that AB and BA have different sizes when $m \neq n$.)

(b) Prove that there are no matrices $A, B \in M_{n\times n}(\mathbb{F})$ with $AB - BA = I$.

(c) Let $W = \operatorname{span}\{AB - BA : A, B \in M_{n\times n}(\mathbb{F})\}$. Prove that

$$W = \{X \in M_{n\times n}(\mathbb{F}) : \operatorname{trace} X = 0\},$$

and find $\dim W$.

[20] 8.

(a) Let $A \in M_{m\times n}(\mathbb{F})$. Prove that if $Ax = 0$ has only the trivial solution, then $A^T y = c$ has a solution for every $c \in \mathbb{F}^n$.

(b) Let $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times p}(\mathbb{F})$. Prove that

$$\operatorname{rank}(AB) \geq \operatorname{rank} A + \operatorname{rank} B - n.$$

(Hint: apply the rank theorem to A restricted to $\operatorname{Ran} B$.)

(c) Let $A \in M_{n\times n}(\mathbb{F})$ satisfy $A^2 = A$. Prove that $\operatorname{rank} A + \operatorname{rank}(I - A) = n$.

[+10] Bonus. Prove that a vector space over $\mathbb{R}$ cannot be written as the union of three proper subspaces. (A subspace $U \subseteq V$ is *proper* if $U \neq V$.)